How to Design a Liquid Rocket Engine

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Abstract

This paper examines the various factors to consider when designing a liquid rocket engine. The primary focus is on the geometry of the engine, including the chamber, nozzle, and throat. Determining the optimal geometry of a rocket engine is essential to maximize efficiency, reduce thermal and vibrational loads, and minimize weight. The following paper provides a step-by-step guide to the design process of a liquid rocket engine.

Nomenclature

Parameters

$\dot{m}$	mass flow rate
γ	specific heat ratio
ρ	density
A	area
a	speed of sound
F	thrust
L^*	characteristic length
M	mach
P	pressure
R	gas constant
T	temperature
V	volume
v	velocity

Subscripts

0	free stream
c	chamber
e	nozzle exit
t	throat

1. Getting Started

Liquid rocket engines operate by forcing a fuel and an oxidizer into a combustion chamber where they are ignited and burn. The hot combustion gases from this process are pushed through the converging section of a nozzle to the throat, where the flow is choked, and the gases travel at exactly Mach 1. Upon exiting the throat, the combustion gases are expanded in the diverging section of the nozzle, where the velocity of the gases increases as the pressure decreases. The acceleration of the combustion gases creates thrust. The geometry of the entire engine, from the chamber to the throat to the nozzle, is extremely important and dictates a significant portion of the engine's efficiency, as well as whether the engine will operate at all. For reference throughout this paper, the variable designations in Figure 1 will be used to denote the various critical dimensions of the rocket engine.

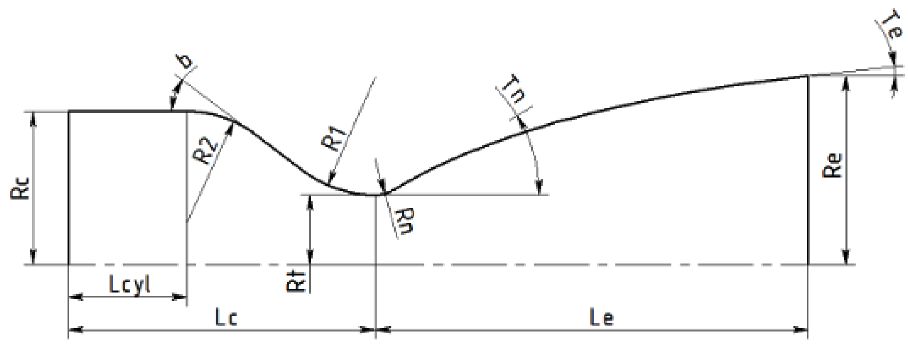


Figure 1: Rocket Engine Geometry

1.1. Propellants

When designing a liquid rocket engine, the first specifications that must be chosen are the fuel and oxidizer to be used in the engine. Choosing the optimal propellants for liquid rocket engines has been the topic of research for decades, but most modern commercial rockets use liquid hydrogen (LH4) as fuel and liquid oxygen (LOX) as oxidizer [5]. However, due to handling difficulties with LH4, it is very rarely used in amateur engines. Fuels for amateur engines are typically ethanol, kerosene, or RP-1. Fuel choice should be based on the ease of handling, safety, and combustion properties. LOX remains the common choice for amateur engines because, despite being cryogenic, it is relatively easy to handle and is extremely efficient in combustion.

1.2. Thrust

After choosing the propellants for the engine, the desired thrust of the engine must be decided. If the engine is being designed for use on a rocket, the thrust must be high enough to counteract the weight of the launch vehicle. Unless thrust vectoring or some other active stability control system is integrated, a thrust-to-weight ratio of at least 5 is recommended for amateur rockets. The magnitude of the thrust will affect the physical size of the engine as well as the propellant mass flow rates.

1.3. Chamber Pressure

The next parameter to decide is the chamber pressure. A higher chamber pressure is desirable because it increases the specific impulse of the engine and creates more thrust for an engine of the same size. However, a high chamber pressure also has the effects of increasing the structural and thermal loads on the chamber, which contributes to a heavier engine and a more complex cooling system. Chamber pressures can range from about 400 psi for amateur engines to several thousand psi for large commercial rockets. Chemical Equilibrium with Applications (CEA) is a software designed at NASA that is very helpful in determining chamber pressure [10].

1.4. Mixture Ratio

Mixture ratio, or O/F Ratio, is the mass flow rate of oxidizer divided by the mass flow rate of fuel in the engine. The mixture ratio is important for dictating the qualities of the combustion process and significantly impacts the temperature of the chamber, the specific impulse of the engine, and the types of combustion gases present. A review of existing literature for the chosen propellant combination, or the use of CEA, is highly recommended to determine the mixture ratio.

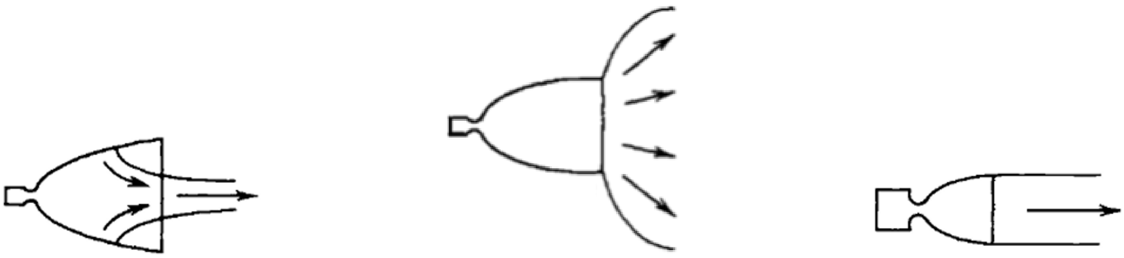
2. Isentropic Equations

After determining the propellants, thrust, chamber pressure, and mixture ratio of the engine, the isentropic flow equations can be used to determine the parameters and geometry for several essential aspects of the engine [11]. Isentropic flow assumes the flow of the combustion gases is both adiabatic (no heat transfer occurs) and reversible. This assumption is not a perfect representation of the flow in reality, but it is sufficiently accurate to provide a preliminary geometry. Because the chamber pressure and atmospheric pressure are known, these values can be used to solve for M_e, the Mach number of the combustion gases as they exit the nozzle. Assuming the rocket is optimized for firing at approximately sea level, P_e, the pressure at the exit of the nozzle should be equal to the atmospheric pressure. This maximizes the efficiency of the engine and prevents over- or under-expansion. Over-expansion occurs when the atmospheric pressure is greater than the pressure of the exit gases. This can cause shockwaves and waste energy that could otherwise contribute to thrust. Under-expansion occurs when the atmospheric pressure is less than the pressure of the exit gases. This is inefficient because the excess pressure could have been used to increase the velocity of the exit gases. Under-expansion is better than over-expansion, but both result in efficiency losses. The effects of over- and under-expansion are shown in Figure 2.

Now that P_e is determined, Eq. (1) can be used to calculate M_e. The specific heat ratio, γ, is based on the propellant combination and mixture ratio and is most easily obtained from CEA.

$$\frac{P}{P_c} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{-\gamma}{\gamma - 1}} \tag{1}$$

The exit mach can be used to determine the area ratio between the exit of the nozzle and the throat. This represents how much the gases must expand in order to reach the desired Mach number by the end of the nozzle. The area ratio can be calculated from Eq. (2).



(a) Over Expansion (b) Under Expansion (c) Full Expansion

Figure 2: Exhaust Gas Expansion Regimes [16]

$$\frac{A}{A_t} = \frac{1}{M}\sqrt{\left[\frac{1+\left(\frac{\gamma-1}{2}\right)M^2}{1+\left(\frac{\gamma-1}{2}\right)}\right]^{\frac{\gamma+1}{\gamma-1}}} \tag{2}$$

The next parameter to find is the mass flow rate of the propellants. In order to calculate this, the conditions at the exit of the nozzle must be found. The first to be calculated is the temperature by using the exit Mach number. The temperature is calculated by Eq. (3).

$$\frac{T}{T_c} = \left(1 + \frac{\gamma - 1}{2}M^2\right)^{-1} \tag{3}$$

The temperature T_e can be used to calculate the speed of sound at the nozzle exit, which is necessary to convert the exit Mach number to the velocity of the combustion gases as they exit the nozzle. These relations are shown in Equations (4) and (5). It should be noted that R is the gas constant of the combustion gases, found by dividing the universal gas constant by the molecular weight of the gas.

$$a = \sqrt{\gamma RT} \tag{4}$$

$$v = aM \tag{5}$$

The thrust equation can be used to calculate the mass flow rate of the propellants now that v_e has been found. The full thrust equation is shown in Eq. (6), but because the velocity of the surrounding air is negligible relative to the velocity of the exit gases, and because the exit pressure is equal to the atmospheric pressure, the equation simplifies to Eq. (7) which can be used to calculate the total mass flow rate. The mixture ratio can then be used to calculate the mass flow rates of the fuel and oxidizer.

$$F = \dot{m}_e v_e - \dot{m}_0 v_0 + (P_e - P_0)A_e \tag{6}$$

$$F = \dot{m} v_e \tag{7}$$

Now the mass flow rate can be used to calculate the area of the nozzle throat using the choked flow equation [3]. Eq. (8) can be used to calculate the throat area. This value can be multiplied by the previously calculated area ratio to determine the area of the nozzle exit.

$$A_t = \frac{\dot{m}}{P_c \sqrt{\frac{\gamma}{RT_c}} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}} \tag{8}$$

The rough shape of the engine is beginning to come together, but the precise geometry of the combustion chamber and nozzle still must be determined.

3. Combustion Chamber

While spherical combustion chambers are technically most efficient, the difficulty in machining them makes them impractical for most rocket engines. Cylindrical combustion chambers are by far the most common shape, and they consist of a cylinder of cylinder that tapers down to the throat area near the end. The characteristic length, L^*, is the parameter usually used to dictate the size of the combustion chamber, and it is most commonly determined from tables in existing literature with similar propellant combinations and chamber conditions. L^* refers to the length the combustion chamber would need to be if the entire chamber were a cylinder with the same radius as the throat of the nozzle. Determining an accurate value for L^* is important because a larger chamber than necessary introduces extra losses due to heat transfer and unnecessary weight. A smaller chamber than necessary leads to incomplete combustion, which contributes to efficiency losses and combustion instabilities [9]. L^* is defined mathematically in Eq. (9), and some typical values for L^* are shown in Table 1. A method to numerically determine L^* is shown in Appendix A.

$$L^* = \frac{V_c}{A_t} \tag{9}$$

Table 1: Typical Chamber Characteristic Length, L^* [4]

Propellant Combination	**L^*, cm**
Nitric acid/hydrazine-base fuel	76–89
Nitrogen tetroxide/hydrazine-base fuel	76–89
Hydrogen peroxide/RP-1 (including catalyst bed)	152–178
Liquid oxygen/RP-1	102–127
Liquid oxygen/ammonia	76–102
Liquid oxygen/liquid hydrogen (GH_2 injection)	56–71
Liquid oxygen/liquid hydrogen (LH_2 injection)	76–102
Liquid fluorine/liquid hydrogen (GH_2 injection)	56–66
Liquid fluorine/liquid hydrogen (LH_2 injection)	64–76
Liquid fluorine/hydrazine	61–71
Chlorine trifluoride/hydrazine-base fuel	51–89

3.1. Converging Section

Once the value for L^* is determined, Eq. (9) can be used to determine the necessary volume of the chamber. Note that this is the total chamber volume, including both the cylindrical and converging sections. When determining the two converging radii, R_2 does not matter significantly and can be anywhere from twice to ten times the radius of the throat. R_1, on the other hand, matters slightly more and is typically set at 1.5 times the radius of the throat to create a smooth transition to the horizontal geometry and prevent any flow separation from occurring [2]. Increasing R_1 and R_2 has the benefit of increased efficiency at the cost of increased chamber length, which increases weight. The convergence angle b should be about 30° because this balances weight savings and flow efficiency. The maximum flow efficiency occurs around 25°, but this significantly increases the length of the chamber with only marginal efficiency gains [15].

The parameters R_1, R_2, and b can be used to calculate the length of the converging section of the chamber. Geometry is used to yield Eq. (10). The length of the converging section is used to calculate the approximate volume of the converging section using the formula for a truncated cone shown in Eq. (11).

$$L_{conv} = R_1 \sin b + R_2 \sin b + [R_c - R_t - R_1 (1 - \cos b) - R_2 (1 - \cos b)] \tan b \tag{10}$$

$$V_{conv} = \frac{1}{3}\pi \left(R_c^2 + R_c R_t + R_t^2\right) L_{conv} \tag{11}$$

3.2. Cylindrical Section

The volume of the converging section can be subtracted from the total chamber volume (derived from L^*) to give the volume of the cylindrical section of the chamber. In order to determine the length of the cylindrical section L_{cyl}, the area of the chamber must be known. This is derived from the contraction ratio, which is the area of the chamber divided by the area of the throat. For large, commercial engines, the contraction ratio must be carefully determined to maximize combustion efficiency and typically ranges from 2 to 5 [9]. However, for smaller, amateur engines, the contraction ratio is usually dictated by the size of the injector and can be as large as 15 to 20.

4. Nozzle

There are three common geometries for the nozzle where the combustion gases expand and accelerate after passing through the throat. Conical nozzles are the simplest shape and consist of an angled straight line extending from the throat to the nozzle exit. While this is the easiest to machine, it is also the least efficient nozzle geometry. Another option is the parabolic nozzle, where the combustion gases expand from the throat to the nozzle exit along a parabolic arc. The parabolic nozzle is significantly more efficient than the conical nozzle while still having a simple analytical solution. The third and most efficient nozzle geometry utilizes the method of characteristics (MOC). MOC is a way to model the Mach lines created from the supersonic flow and ensure that when each Mach line intercepts the nozzle wall, it is redirected to flow axially out of the nozzle exit. This ensures all of the thrust is being directed axially out of the rocket engine, which leads to maximum efficiency. MOC requires significantly more modeling than the conical or parabolic nozzles, and the geometry can only be found numerically.

Regardless of the nozzle methodology chosen, it is recommended to incorporate a small transition radius from the throat to the beginning of the nozzle. [7] suggests a transition radius of 0.382 times the throat radius. The transition radius helps prevent flow separation and decreases the heat loading at the transition point.

4.1. Conical Nozzle

A conical nozzle typically consists of an initial transition radius (R_v) followed by a linear section extending to the nozzle exit. A typical value for R_v is about 1.5 times the throat radius [7]. The radius sweeps out an angle, α, which is the same as the angle of the linear section with respect to the centerline of the nozzle. The total length of the nozzle is represented by Eq. (12).

$$L = \frac{R_t}{\tan\alpha}\left[\frac{R_e}{R_t} - 1 + \frac{R_v}{R_t}(\sec\alpha - 1)\right] \tag{12}$$

A larger α decreases the length of the nozzle but comes at the cost of lower thrust. Assuming spherical expansion, the actual thrust level can be estimated by a correction term where $F = \lambda \dot{m} v_e$, where λ is dependent on the expansion angle and is expressed by Eq. (13) [7].

$$\lambda = \frac{1 + \cos\alpha}{2} \tag{13}$$

4.2. Parabolic Nozzle

The optimal parabolic curve can be approximated using the endpoint conditions given by the isentropic flow equations, an initial expansion angle, and the nozzle length. The initial expansion angle was calculated by the Prandtl–Meyer function shown in Eq. (14). This function is used to determine the angle that a gas traveling sonically transitions isentropically to traveling supersonically. The initial expansion angle of the nozzle from the horizontal is given by ½ the Prandtl-Meyer angle. The nozzle length was calculated by finding the length a 15° conical nozzle would need to be to achieve the area ratio. The length of a parabolic nozzle, L_n, is typically chosen to be between 70% and 100% the length of a 15° conical nozzle [17]. A parabolic approximation was used to determine the geometry of the nozzle using the initial angle, the endpoints of the nozzle, and the nozzle length.

$$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}} \cdot \arctan\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} - \arctan\sqrt{M^2-1} \tag{14}$$

The parabolic curve for the nozzle wall can be found by using a parabola in the form $y = \sqrt{Ax+B} + C$ where A, B, and C are initially unknown. Equations 15 and 16 are generated by plugging in the two endpoints into the equation of the parabola. Equation 17 is found by setting the derivative of the parabolic equation $\left(\frac{dy}{dx} = \frac{A}{2\sqrt{Ax+B}}\right)$ equal to the tangent of the maximum expansion angle. The transition radius is denoted as R_v. For reference, [17] suggests that the initial expansion angle should be between 28 and 30 degrees, and the expansion angle at the nozzle exit should be between 10 and 14 degrees.

$$R_t + R_v \left[1 - \cos\left(\frac{1}{2}\nu(M_e)\right)\right] = \sqrt{AR_v + B} + C \tag{15}$$

$$R_e = \sqrt{AL_n + B} + C \tag{16}$$

$$\frac{A}{2\sqrt{AR_v \sin\left(\frac{1}{2}\nu(M_e)\right) + B}} = \tan\left(\frac{1}{2}\nu(M_e)\right) \tag{17}$$

4.3. Method of Characteristics Nozzle

The state of a supersonic fluid is defined by its Mach number (M) and flow angle (θ). The propagation of disturbances in such a flow occurs along characteristic curves. For isentropic expansion, the relationship between the Mach number and the turning angle of the flow is given by the Prandtl-Meyer function, expressed earlier in Eq. (14).

4.3.1. Characteristic Lines and Invariants

In a two-dimensional supersonic flow, two types of characteristic lines exist, defined by the local Mach angle $\mu = \sin^{-1}\left(\frac{1}{M}\right)$ [13]. The characteristic lines through a point and their associated angles are shown in Figure 3:

- C_- **Characteristics (Left-running):** $\frac{dy}{dx} = \tan(\theta + \mu)$
- C_+ **Characteristics (Right-running):** $\frac{dy}{dx} = \tan(\theta - \mu)$

Along these lines, the quantities $K_\pm = \theta \pm \nu$ are the Riemann invariants of the flow. For a two-dimensional (planar) nozzle, these invariants are exactly constant along their respective characteristics:

$$K_- = \theta + \nu = \text{constant along } C_- \tag{18}$$

$$K_+ = \theta - \nu = \text{constant along } C_+ \tag{19}$$

Rocket nozzles, however, are axisymmetric rather than planar, which changes the mathematics slightly. Because an axisymmetric flow field depends on the radial distance y from the centerline as well as on x, $K_\pm$ are no longer exactly constant along the characteristics; instead, they drift according to

$$dK_- = -\Gamma_-\, dx \text{ along } C_-, \qquad dK_+ = \Gamma_+\, dx \text{ along } C_+ \tag{20}$$

where $\Gamma_\pm$ depend on the local Mach number, flow angle, and radial position [18]. Physically, $\Gamma_\pm$ accounts for the fact that an axisymmetric flow must expand in two transverse directions rather than one, so the same amount of turning draws on more streamtube area near the centerline than far from it. This correction vanishes as $y \to \infty$, recovering the planar result above, and grows as the flow turns closer to the axis – exactly the region a rocket nozzle's expansion fan passes through.

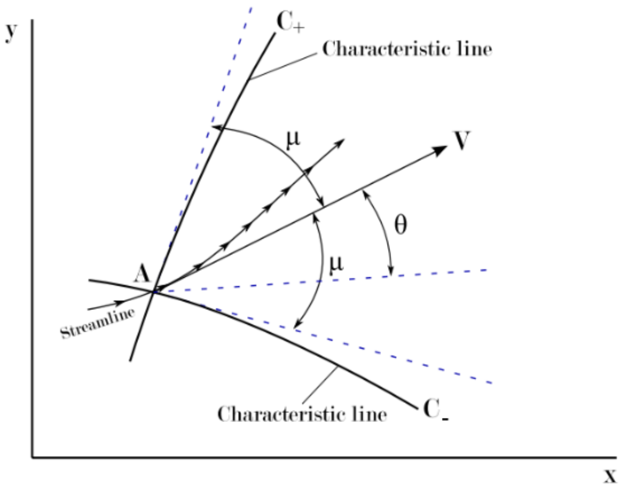


Figure 3: Left- and right-running characteristic lines [6]

4.3.2. Numerical Methodology

The numerical implementation represents the flow field as a discrete grid of n characteristic lines (see Figure 4). A larger n increases the accuracy of the solution but increases computational cost. The flow is sonic ($M = 1$) at the throat. A series of expansion waves is generated at the throat corner, with the maximum expansion angle θ_{max} setting the angle of the final expansion wave. The angle of each initial expansion wave is interpolated from 0 to θ_{max}. This ensures that the final wave reflection results in a flow angle of $\theta = 0$ at the nozzle exit.

For a planar nozzle, θ_{max} has a closed-form solution equal to half the Prandtl-Meyer angle of the exit Mach number, $\theta_{max} = \frac{1}{2}\nu(M_e)$. Because $K_\pm$ are not conserved along the characteristics of an axisymmetric flow, this shortcut no longer holds exactly: the θ_{max} that produces the desired exit Mach number must instead be found iteratively, by solving a trial mesh, checking the resulting exit Mach number, and adjusting θ_{max} until the two agree. For a nozzle with a modest overall turning angle, the planar expression above is a good starting guess for this iteration. For a nozzle that turns the flow sharply, however, the iteration is instead built up gradually: starting from a small, easily-solved θ_{max} and stepping it up toward the target, warm-starting each new mesh from the last converged one. This continuation strategy keeps the iterative solver well-behaved even for aggressively expanding nozzle designs, where attempting to solve the full turning angle directly can fail to converge.

For any interior point where a C_- line and a C_+ line intersect, the local properties are found from the invariants carried in from the upstream points [13], corrected for the axisymmetric drift of Eq. (20) over the short segment connecting each upstream point to the new point. Once the corrected $K_\pm$ are known at the new point, Eq. (14) can be used to solve for the Mach number at that point, which can be used to find μ.

$$\theta = \frac{1}{2}(K_- + K_+) \quad \text{and} \quad \nu = \frac{1}{2}(K_- - K_+) \tag{21}$$

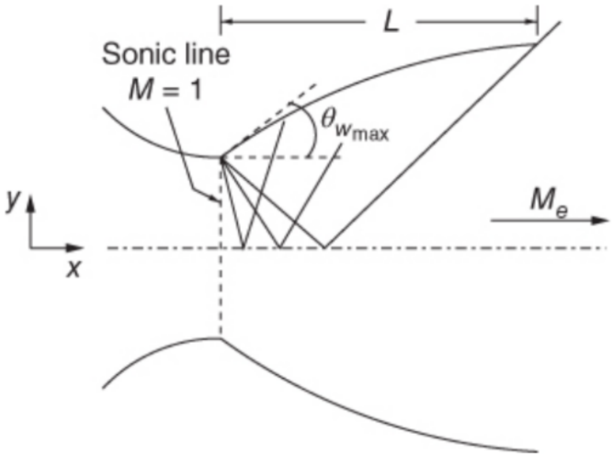


Figure 4: Discretized flow field in MOC nozzle [18]

Two boundary conditions are enforced:

1. **Centerline:** The flow angle θ is set to zero. This effectively reflects waves, but in actuality represents the flow lines from the opposite side of the centerline crossing over. This simplification works due to the axial symmetry of the nozzle.
2. **Wall:** The wall slope must match the local flow angle to reflect the waves axially out of the engine.

The flow characteristics can be used to find the spatial (x,y) coordinates of the intersection points of the left and right running characteristic lines. The slopes of the left and right running characteristic lines are based on the average of the flow angles from the previous and following intersection points and are calculated in Equations (22) and (23), respectively [13]. A and B are the previous points on the left and right running characteristics, and P is the following point where both lines intersect.

$$m_- = \tan \frac{(\theta_A - \mu_A) + (\theta_P - \mu_P)}{2} \tag{22}$$

$$m_+ = \tan \frac{(\theta_B + \mu_B) + (\theta_P + \mu_P)}{2} \tag{23}$$

These slopes can be used with the coordinates of the previous points to find the coordinates of the new point, P [13].

$$x_P = \frac{y_A - y_B + m_+ x_B - m_- x_A}{m_+ - m_-} \tag{24}$$

$$y_P = y_A + m_-(x_P - x_A) \tag{25}$$

After calculating the coordinates of all the points and extracting the coordinates of the wall points, a cubic interpolation can be used to "smooth" the spline formed by the wall points. A smooth nozzle geometry is necessary to ensure the flow remains isentropic and expands efficiently.

The MOC approach to nozzle design is the only method discussed that does not use the area ratio calculated from Eq. (2). For nozzles with a modest overall turning angle, the area ratio derived from the MOC and the area ratio calculated from the isentropic equation are very similar; [12] found a difference of less than 1% between the two methods. This gap grows, however, for nozzles that turn the flow more sharply, since it is driven by the same axisymmetric effect discussed above rather than by numerical error, and can reach several percent for aggressively expanding designs. A nozzle designed with the MOC should therefore be checked against its own calculated area ratio rather than assumed to match the isentropic value.

It should be noted that the MOC discussed in this section assumes a sharp transition from the horizontal throat to the Prandtl-Meyer angle. As previously mentioned, it is best practice to have a small transition radius between these sections. There is some literature on how to adapt the MOC to account for the addition of this radius, but it makes the algorithm considerably more difficult to implement, and if the transition radius is small, it has a negligible effect on the wall geometry.

Appendix A. Calculating Characteristic Length (L^*)

There are several models for calculating L^*. Due to its accuracy, the Adler model [1] was chosen because it accounted for the length over which chemical reactions take place during the combustion process. This contributes to a slightly longer L^* than that calculated by the Spalding model, which assumes chemical loading tends to zero and that chemical reactions occur in an infinitesimal length. The first step in the process is determining the size of the fuel droplets after the injected fuel is atomized. The Sauter Mean Diameter (SMD) of the droplets measures the average diameter of a given droplet, and the calculation of SMD depends heavily on the injection method used. For a pressure swirl injector, the Radcliffe model [14] can be used to calculate SMD by Eq. (A.1), where σ is the surface tension of the liquid, μ is the dynamic gas viscosity, $\dot{m}_l$ is the mass flow rate of the liquid fuel, and ΔP is the pressure drop over the injector.

$$d_{32} = 7.3\sigma^{0.6}\mu^{0.2}\dot{m}_l^{0.25}\Delta P^{-0.4} \tag{A.1}$$

For an impinging injector, the Ryan-Anderson Model [19] can be used. This model for SMD is represented by Eq. (A.2), where d_j is the diameter of the impinging jets, C is the ratio of gas to liquid density, We is the Weber number of the fuel, v_i is the injection velocity, and θ is the impingement half-angle.

$$d_{32} = \left(\frac{2.62}{64^{1/3}}\right) d_j C^{-1/6} \left[We * f(\theta)\right]^{-1/3} \tag{A.2}$$

$$C = \frac{\rho_g}{\rho_l} \tag{A.3}$$

$$We = \frac{\rho_l v_i^2 d_j}{\sigma} \tag{A.4}$$

$$f(\theta) = \frac{(1 - \cos\theta)^2}{\sin^3\theta} \tag{A.5}$$

$$v_i = \sqrt{\frac{2\Delta P}{\rho_l}} \tag{A.6}$$

The Adler model proposes the use of dimensionless parameters for droplet radius, velocity, and temperature. It also utilizes the parameter η, which is the fractional decrease in droplet radius and is calculated by Eq. (A.7), where $r_0 = d_{32}/2$.

$$\eta = 1 - \frac{r}{r_0} = 1 - \zeta \tag{A.7}$$

The differential equation for droplet velocity is defined in Eq. (A.8) where $\chi = v/v_g$, the droplet velocity relative to the final combustion gas velocity, and S is the droplet drag. The differential equation for droplet temperature is defined in Eq. (A.9), where $\tau = T/T_g$, the temperature relative to the final combustion gas temperature, where ψ is a dimensionless reaction rate function calculated by Eq. (A.10), and where L is a chemical loading. n is an integer that modifies the form of the reaction rate, c_p is the specific heat at constant pressure of the gas, k is the average gas conductivity, B is the transfer number, H is the fuel calorific value, m_o is the weight concentration of oxygen, Q is the vaporization enthalpy, z is the weight of oxygen required for combustion of unit weight fuel, $c_{p,l}$ is the specific heat at constant pressure of liquid fuel, and T_s is the boiling temperature of liquid fuel.

$$\frac{d\chi}{d\eta} = \frac{S}{(1-\eta)\tau}\left[\eta(\eta^2 - 3\eta + 3) - \chi\right] \tag{A.8}$$

$$\frac{d\tau}{d\eta} = \frac{1-\eta}{\eta(\eta^2 - 3\eta + 3)}\left[\frac{\psi}{L\tau}\chi - 3(1-\eta)\tau\right] \tag{A.9}$$

$$\psi(\tau) = (n+1)\left(1 + \frac{1}{n}\right)^n (1-\tau)\tau^n \tag{A.10}$$

$$S = \frac{9c_p\mu/k}{2\ln(1+B)} \tag{A.11}$$

$$B = \frac{Hm_0}{Qz} + c_{p,l}\frac{T_g - T_s}{Q} \tag{A.12}$$

Because the length of the chemical reactions is being considered in the Adler model, the chemical loading term no longer tends to zero, and is calculated by Eq. (A.13). χ_0 is calculated by dividing the initial velocity upon injection $v_i = \sqrt{\frac{2\Delta P}{\rho_l}}$ [8] by the final gas velocity which is determined from the isentropic area-mach relation in Equations (2) and (5). L varies from 0 to a critical value L_c, calculated from Eq. (A.14). If $L < L_c$, combustion is possible in the chamber.

$$L = \frac{\chi_0 * \psi(\tau_0)}{3\tau_0^2} \tag{A.13}$$

$$L_c = \frac{\chi_0}{3}\left(\frac{n+1}{n-1}\right)\left(1 + \frac{1}{n}\right)^n\left(1 + \frac{1}{n-1}\right)^{n-2} \tag{A.14}$$

Finally, L^* can be calculated by evaluating the integral in Eq. (A.15) and substituting the value into Eq. (A.16) [20]. The boundary conditions for Eq. (A.16) are $\zeta = 0$, $\chi = 1$, $v = v_g$, and $\tau = \tau_1 \leq 1$.

$$\xi^* = \int_0^1 \frac{\chi}{\tau}\zeta \, d\zeta \tag{A.15}$$

$$L^* = \frac{\xi^* a r_0^2\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}\left(\frac{G}{\rho_g a}\right)^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}}{\frac{k}{c_p\rho_l}\ln(1+B)} \tag{A.16}$$

In order to solve the differential equations in Equations (A.8) and (A.9), a Runge-Kutta numerical method can be used. The first step of the numerical solution is calculated analytically by

Equations (A.17) and (A.18), where h is the integration step size. The boundary conditions for Equations (A.17) and (A.18) are $\eta = 0, \chi = \chi_0 = v_0/v_g$, and $\tau = \tau_0$.

$$\chi_1 = \chi_0 + \left(\frac{d\chi}{d\eta}\right)_0 h \tag{A.17}$$

$$\tau_1 = \tau_0 + \left(\frac{d\tau}{d\eta}\right)_0 h = \tau_0 + \left[\frac{(s-\tau_0)(1-\tau_0)}{(n-2)(1-\tau_0)-1}\right] h \tag{A.18}$$

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